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Conference Paper · November 2022

DOI: 10.1109/ITED56637.2022.10051564

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Comparative Analysis of the Performance of Various Support Vector Machine kernels

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Abstract: Support Vector Machine (SVM) in dealing with a classification problem, separates classes using decision boundaries with the primary objective of establishing a large margin between support vectors of the respective class groups; it utilizes kernels to achieve non-linear decision boundaries. This current work examines the performance of four SVM kernels (Sigmoid, Linear, Radial Basis Function (RBF) and Polynomial kernel functions) in addressing classification problems using two datasets from two domains. The two datasets are the Knowledge Discovery in Dataset (KDD) and a set of features extracted from voiced and unvoiced frames. The Polynomial kernel function had the best classification performance on the KDD dataset with accuracy and precision of 99.77% and 99.8% respectively but recorded the worst performance against the voice-feature dataset with an accuracy of 74.96%. Inductively, the polynomial kernel can be best suited for some classification datasets but can return the worst classification performance on another classification dataset. The RBF shows consistent high performance across the two data domains with accuracies of 96.04% and 99.77% and can be considered a general-purpose kernel guaranteed to yield satisfactory classification performance regardless of the dataset type or data domains. The performance of polynomial kernels over the two separate datasets supports the "No Free Lunch Theorem", which when applied to machine learning, means that if an algorithm performs well over a class of problem, it may have worse performance on other class of problem. This implies that there might not be one specific machine learning algorithm that gives the best possible performance for a set of problems, it is therefore important for researchers to try out various algorithms before concluding on the best possible result on any dataset.

Keywords: Support Vector Machines Kernels, Comparative Analysis, Classification problem, Machine learning.

I. INTRODUCTION

Artificial Intelligence (AI) is about developing intelligent systems that can do tasks that are normally performed by humans and one of the branches of AI is machine learning which deals with the development of programs that perform tasks though not being explicitly programmed, whose performance on the job improves as more samples are provided. Techniques in machine learning are categorized into three, supervised learning, unsupervised learning, and reinforcement learning. Of interest to this research is supervised learning where input features and corresponding class labels/output are provided from which the supervised algorithm learns patterns that connect the input and output resulting in models used in predicting unknown input samples[1].

Among several algorithms of supervised learning is the support vector machine (SVM) which was introduced by Boser, Guyon and Vanik in 1992 [2]. SVM is used for both classification and regression problems. It separates classes using hyperplane as the class boundary otherwise known as the decision boundary [3] and the main objective of SVM is to maximize the distance between the closest data points from the different classes to the decision boundary called Support Vectors hence it is called large margin machine or optimal margin classifier and this guarantee that unknown samples are accurately classified[4].

SVM while maximizing the separation margin between classes automatically prevent the over-fitting of data, so it does generalize well. SVM is easy to train, and it avoids local optimal, a challenge experienced by the neural network. In terms of performance at classification problems, the result obtained from SVM can be at par with neural network and sometimes better as reported by [5] in their work where neural network achieved a classification accuracy of 78.53% while SVM achieved 98.25% in distinguishing between infected red blood cells. Similarly,[6] in their design and Implementation of a voice activity detection system made use of SVM as the classifier for

distinguishing between voiced and unvoiced frames, achieving a classification accuracy of 100% with aid of a feature selector, a performance that more complex neural network architectures did not achieve on similar projects. There are other reported research works which show that Artificial Neural Networks(ANNs) outperformed SVM, among them is the literature reviews done by [7] where ANN recorded 99.71% accuracy while SVM achieved 85% for fraudulent transaction detection.

Some of the good qualities of SVM are: (1) It handles overfitting, outliers, and noisy data completely well, (2) It can deal with scenarios where the number of variables is more than training examples comfortably and (3) It automatically detects nonlinearity in the data, hence, transforming the features directly is not required[8]. Support Vector Machine has been used extensively in classification problems and some of the application areas are: Intrusion Detection, Face Analysis, Hand Writing Analysis and so forth [9],[10].

The strength of the SVM lies in the kernel trick that uses a kernel function to transform feature space into a high dimensional and complex space which leads to a nonlinear hyperplane as a decision boundary for class separation without resulting in computation cost overhead and increasing the memory demand[8]. Some of the SVM kernel functions are: (1) Linear, (2) Polynomial (3) Gaussian function (4) Radial Basis Function (5) Exponential Radial Function (6) Sigmoid Function [8][9].

The objective of this research is to perform a comparative analysis on the performance of four SVM kernels which are the linear, polynomial, radial basis function and the sigmoid function with two different types of datasets, the KDD and extracted features from voiced and unvoiced frames [6] datasets using standard classification metrics for the evaluation.

II. REVIEW OF RELATED WORKS

Support Vector Machines (SVM) are powerful but flexible supervised machine learning algorithms which are used both for classification and regression. It is also one of the most efficient algorithms used for smaller datasets. [11] Support Vector Machines and their variants and extensions, often called kernel-based methods (or simply kernel methods), have been studied extensively and applied to various pattern classification and function approximation problems [12].

[13] provided a summary of Support Vector Machines and kernel functions; constructed experiments with various benchmark datasets to compare the performance of various kernel functions. The outcome, based on the cosine-score, related to

accuracy and the standard deviation of the 10-fold-validation, showed that it was effective to employ kernel functions for general datasets, such as ANOVA kernel, Gaussian kernel, rational quadratic kernel, log kernel, Laplacian kernel, histogram intersection kernel, Vovks real polynomial kernel, and inverse multi-quadratic kernel.

[14] compared the performance of linear and polynomial kernel functions in sentiment classification with the YouTube comments dataset on the COVID-19 pandemic in Indonesia. The polynomial kernel had an accuracy of 91%, RBF kernel 90% and linear kernel 86%. Specificity also exhibited the same trend. In terms of sensitivity, the polynomial kernel had the worst performance of 20%, and the best performance of 64% came from the linear kernel

[15] investigated the optimal parameter by comparing the different kernel functions in order to implement an SVM classifier. The NSL-KDD dataset in a highly imbalanced form was used to test the model for robustness. The outcome showed that the model of SVM with optimum parameters can attain 99.9% on the NSL-KDD dataset.

[16] evaluated six different kernels by varying the training ratio in the proportion of 60-20-20, 40-30-30 and 20-40-40 to investigate the performance. The outcome showed that the SVM model with Gaussian RBF and Exponential RBF kernels are the best suited for call algorithm data in terms of best specificity and sensitivity values. The polynomial and linear kernels have worse performance.

Performance evaluation of four SVM kernels was carried out on three separate datasets of Mushroom, SCD, and chronic kidney disease datasets. The RBF and linear kernels reported competitively high classification accuracies across the three datasets ranging from 99% to 100%. The performance of the sigmoid kernel was also encouraging with accuracy in the range of 97.2% to 99.3%. With regards to the polynomial kernel, it recorded high classification accuracies of 98.8% and 99% for SCD and Chronic Kidney Disease datasets respectively but had poor accuracy of 54.4% on the Mushroom dataset[17]

[18] investigated the performance and carry out a comparative analysis of Support Vector Machine (SVM) based classifiers namely SMO-C, C-SVM-C, and NU-SVM-C using public datasets. The outcome showed that the performance of SMO-C is more promising than the other SVM-based classifiers.

[19] focused on a detailed survey about the different algorithms applied to big data analytics and did a comparative study of SVM and SSVM. It was discovered that the sequential SVM is difficult to work with on a large-scale dataset unlike when the smoothing technique is applied to the dataset.

[20], in building breast cancer prediction models, the result of their research, showed that ensemble based on linear kernel using bagging technique and ensemble of bagged RBF SVM had better performance on small scale dataset. For large dataset ensemble of SVM with RBF kernel using boosting method had the best performance.

In comparing the accuracy of anomaly host-based intrusion detection systems,[10] experimented with SVM using Polynomial, Radial Basis Function (RBF), Linear Function and Sigmoid function as the kernel functions. RBF-based intrusion detection reported the best performance with an accuracy of 82.6869%, then followed by the linear kernel with Sigmoid at the bottom of the chart.

III. SVM MATHEMATICAL BACKGROUND

The decision boundary function as a separator which divides the data points into their respective classes, where the support vectors are the data points closest to the boundary. Considering a training sample (x_i, y_i) where the class label is y_i and the feature is x_i , $y \in \mathbb{R}$ ($y \in \{1, -1\}$) and $x \in \mathbb{R}^N$. Figure 1 shows the hyperplanes of support vectors and decision boundaries.

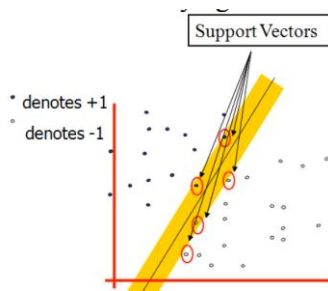


Fig. 1a. SVM Support Vectors[9]

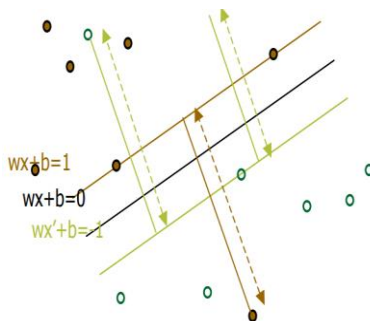


Fig. 1b. SVM hyperplanes [9]

For the positive class $y = 1$, the linear equation is given by equation (1), likewise, for the negative class $y = -1$, is represented by equation (2)

$$wx_i + b \geq 1 \quad (1)$$

$$wx_i + b \leq -1 \quad (2)$$

The hyperplanes of the support vectors are expressed with equations (3) and (4).

$$wx^+ + b = 1 \quad (3)$$

$$wx^- + b = -1 \quad (4)$$

The distance separating the support vectors of the two classes is expressed as equation (5) which is obtained by subtracting equation (4) from equation (3)

$$\text{Margin of separation} = \frac{2}{w} \quad (5)$$

The objective of SVM is to maximize the separation margin:

$$\text{Maximizing Margin} = \frac{2}{||w||} \quad (6)$$

Equation (6) can be expressed as a minimization problem with equation (7).

$$\text{Minimize} \quad \frac{||w||^2}{2} \quad (7)$$

For correct classification of the samples yield equation (8) which refers to a perfect separable dataset (where $y_i(wx_i + b)$ is the functional margin

$$y_i(wx_i + b) \geq 1 \quad (8)$$

The combination of equations (7) and (8) result in a constraint optimization problem below:

$$\text{Minimize} \quad f(w, b) = \frac{||w||^2}{2}$$

$$\text{Subject to } g(w, b) = y_i(wx + b) \geq 1, i = 1, \dots, n$$

The above constraint optimization problem will be solved using Lagrange Multipliers stated in equation (9)

$$\nabla f(x) = \lambda \nabla g(x) \quad (9)$$

Equation 9 simply means that the gradient of function $f(x)$ is equal to a multiple of the gradient of function $g(x)$ where λ is the Lagrange multiplier. In the practical sense, if function $f(x)$ is constrained by function $g(x)$, the extremum is obtained when the normal of these functions are tangential to each other.

With equation (10), Lagrangian is introduced, whose gradient yields equation (11) based on equation (9)

$$L(x, \lambda) = f(x) - \lambda g(x) \quad (10)$$

$$\nabla L(x, \lambda) = \nabla f(x) - \lambda \nabla g(x) = 0 \quad (11)$$

Representing Lagrangian in equation (10) to account for multiple constraints yields equation (12).

$$L(x, \lambda) = f(x) - \sum_i \lambda_i g_i(x) \quad (12)$$

Applying the Lagrange multiplier to the constraint optimization problem of our SVM result into equation (13).

$$L(w, b, \lambda) = \frac{\|w\|^2}{2} - \sum_i \lambda_i [y_i(wx + b) - 1] \quad (13)$$

Substituting equation (13) in (11) we arrive at:

$$\nabla_w L(w, b, \lambda) = \nabla_b L(w, b, \lambda) = \nabla_\lambda L(w, b, \lambda) = 0 \quad (14)$$

Therefore

$$\nabla_w L(w, b, \lambda) = w - \sum_{i=1}^n \lambda_i y_i x_i \quad (15)$$

Which implies

$$w = \sum_{i=1}^n \lambda_i y_i x_i \quad (16)$$

With the derivative with regards to b , we obtained:

$$\nabla_b L(w, b, \lambda) = \sum_{i=1}^n \lambda_i y_i = 0 \quad (17)$$

Substituting the value of w in equation (16) into Lagrangian equation (13) and simplifying, we obtain

$$L(w, b, \lambda) = \sum_i \lambda_i - \frac{1}{2} \sum_i y_i y_j \lambda_i \sum_i y_i y_j \lambda_i \lambda_j x_i x_j - b \sum_i \lambda_i y_i \quad (18)$$

Equation (18) can be simplified into (19) by considering equation (17)

$$L(w, b, \lambda) = \sum_i \lambda_i - \frac{1}{2} \sum_i y_i y_j \lambda_i \sum_i y_i y_j \lambda_i \lambda_j x_i x_j \quad (19)$$

Combining equations (17) and (19) with the constraint of $\lambda_i \geq 0$ yields a dual optimization problem expressed below:

$$\max_{\lambda} W(\lambda) = \sum_i \lambda_i - \frac{1}{2} \sum_i y_i y_j \lambda_i \sum_i y_i y_j \lambda_i \lambda_j x_i x_j$$

Subject to: $\lambda_i \geq 0, i = 1, \dots, n$

$$\sum_i \lambda_i y_i = 0,$$

By solving the maximization problem above, that is finding the λ that maximize $W(\lambda)$ subject to the constraints, then substitute this value in equation (16) to obtain optimal w . From model equation (20), b is extracted

$$b = y_i - wx_i \quad (20)$$

Kernel map input data into complex and high dimensional space that enables SVM to solve non-linear classification problems and this is done using a shortcut that avoids complex computation. The mapping is done by kernel function which transforms input $x \in R^N$ to $\phi(x)$. Some of the kernel functions are listed here.

Linear Kernel Function:

$$k(x_i, x_j) = (x_i \cdot x_j + c)$$

Basic dot product plus a constant c .

Polynomial Kernel Function:

$$k(x_i, x_j) = (x_i \cdot x_j)^d$$

For homogenous polynomial function, d is the degree of the polynomial.

$$k(x_i, x_j) = (x_i \cdot x_j + c)^d$$

For inhomogeneous polynomial function, c is a constant.

Gaussian Kernel Function:

$$k(x_i, x_j) = e^{\frac{-\|x_i - x_j\|^2}{2\sigma^2}}$$

Gaussian Radial Basis Function (RBF):

$$k(x_i, x_j) = e^{(-\gamma \|x_i - x_j\|^2)}, \text{ where } \gamma > 0$$

Laplace RBF kernel Function:

$$k(x_i, x_j) = e^{\frac{-\|x_i - x_j\|}{\delta}}$$

Hyperbolic Tangent Kernel Function:

$$k(x_i, x_j) = \tanh(Kx_i \cdot x_j + c)$$

Sigmoid Kernel Function:

$$k(x_i, x_j) = \tanh(ax_i^T x_j + c)$$

Other kernel functions include the Bessel kernel function, ANOVA radial kernel function, Linear Splines Kernel function, Graph Kernel function, Tree kernel function etc.

With kernel function, the SVM decision boundary equation now becomes

$$y = w\phi(x) + b = 0 \quad (20)$$

IV. METHODOLOGY

Two domain datasets are considered for this performance comparison analysis among the four kernels, the first one the fused time and frequency domain features used by [6] for discriminating between voiced and unvoiced frames of speech signals. The second is the preprocessed KDD datasets for network intrusion detection.

The steps adopted in our methodology are in two major parts as highlighted as follows:

1. The dataset obtained from the speech signal, (each data point represents a voice frame) is divided into two, the training and evaluation datasets in the ratio of 70% to 30% respectively. The various SVM kernels are trained using the training dataset and their respective classification performance against the test dataset is obtained.
2. The preprocessed KDD dataset is also divided into the training and validation datasets in a ratio of 70% to 30% in favour of the training set. The various SVM kernels are trained and their respective classification performance against the test dataset.

The performance metrics to be used are accuracy, precision, recall and F1-score.

The voice signal dataset comprises nineteen features and a class label. The features are standard deviation, skewness, kurtosis, zero-crossing, energy-envelope, root-mean-square energy and thirteen Mel frequency Cepstral Coefficients (Mfcc0 to Mfcc12). This dataset is secondary data obtained from [6]. The total data points in this dataset are 314.

The second dataset is the KDD 99 dataset which is Transmission Control Protocol(TCP) dump extensively used for intrusion detection research[21]. The dataset consists of 42 fields among them are the protocol (TCP, UDP and ICMP), service type and class label. With the removal of duplicates, the total data points dropped from 494,021 to 145,585 which were then preprocessed.

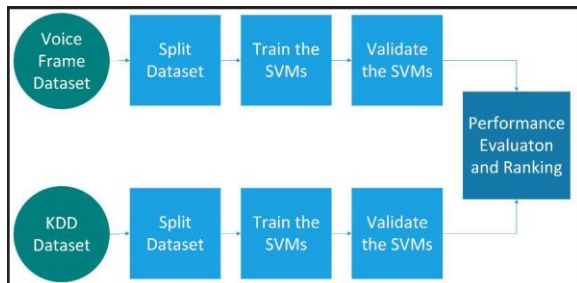


Fig.2. Methodology adopted

The diagram in figure 2 depicts the entire approach adopted as our methodology for this research. For the evaluation process, the ranking of each kernel is considered across the two different data domains.

V. RESULT AND FINDINGS

The training set of the KDD dataset comprised of 101,909 records was used to train SVMs with four respective kernel functions (RBF, Polynomial, Linear and the Sigmoid function), and the test set with 43,676 records was used for the validation exercise. Table I and II are the confusion matrices for RBF and Sigmoid kernel SVMs.

Regarding the speech frame dataset, 95 records which are the test set were used to validate the four trained SVM models with the four respective kernels and the confusion matrixes of Polynomial and Linear kernels are represented in Tables III and IV.

TABLE I. SVM-RBF CONFUSION MATRIX OF KDD DATASET

		PREDICTION	
		FALSE	TRUE
ACTUAL	FALSE	17,364	62
	TRUE	39	26,211

TABLE II. SVM-SIGMOID CONFUSION MATRIX OF KDD DATASET

		PREDICTION	
		FALSE	TRUE
ACTUAL	FALSE	15,315	2,111
	TRUE	2,041	24,209

TABLE III. SVM-POLYNOMIAL CONFUSION MATRIX ON SPEECH FRAME DATASET

		PREDICTION	
		FALSE	TRUE
ACTUAL	FALSE	52	2
	TRUE	29	12

TABLE IV. SVM-LINEAR CONFUSION MATRIX ON SPEECH FRAME DATASET.

		PREDICTION	
		FALSE	TRUE
ACTUAL	FALSE	54	0
	TRUE	0	41

From the confusion matrixes obtained based on the validation outcome of the four respective SVM kernels with the test dataset from the KDD and Speech datasets, four evaluation metrics were extracted, and they are precision, recall, F1-score and accuracy. The values of performance metrics for each of the kernels are presented in figure 3a and 3b.

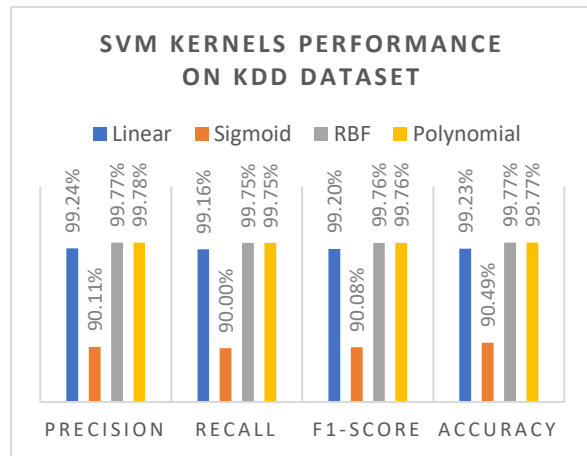


Fig.3a. Performance metrics of the four kernels on KDD Datasets

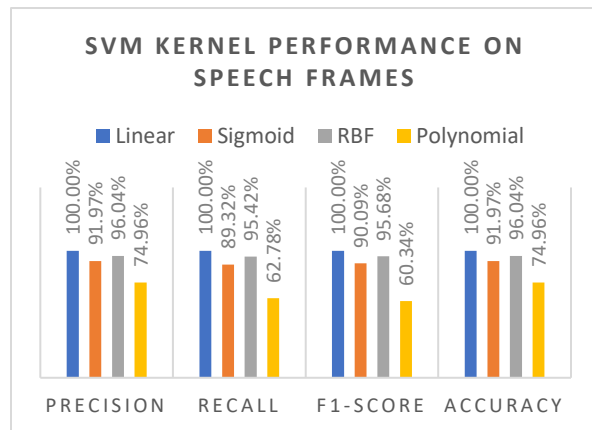


Fig. 3b. Performance metrics of the four kernels on Speech Frame Datasets

From figure 3, Polynomial and RBF-based SVMs had the best performance where they both return 99.75%,99.76% and 99.77% for the recall, F1-score, and accuracy. Their precision performance differs slightly; RBF recorded 99.77% while Polynomial had 99.78%. Linear Kernel ranked 3rd while Sigmoid ranked last in terms of performance across the four metrics.

Concerning Speech Frame Dataset, the Linear kernel that was ranked 3rd and on the KDD dataset was ranked 1st recording 100% for the precision, recall, F1-score and accuracy metrics. RBF kernel ranked 2nd with precision, recall, F1-score, and accuracy of 96.04%,95.42%,95.68% and 95.79% respectively. The polynomial kernel that topped the chart on the KDD dataset is now at the bottom using the Speech Frame Dataset. RBF has shown consistent high performance across the two different domain sets.

VI. CONCLUSION

The Polynomial kernel had the best classification performance with the KDD dataset but recorded the worst performance among the four kernels with precision, recall, F1-score, and accuracy of 74.96%,62.78%,60.34% and 67.37% respectively with regard to the speech frame dataset. It can be induced, that the polynomial kernel is most suitable for some classification problem-based datasets type but can return the worst classification performance on another classification dataset. The RBF shows consistent high performance across the two data domains and can be considered as a general-purpose kernel guaranteed to yield satisfactory classification performance regardless of the dataset type or data domains. The kernel recorded an accuracy of 99.77% on the KDD dataset and the Linear kernel recorded 99.23%. The linear kernel can provide incontestable high performance depending on the type of dataset, as demonstrated in the case of the dataset extracted from speech, where it yielded 100% for accuracy, precision and F1-score; it is also important to note that its worst performance on other datasets is still competitive with best-performing kernels on the other dataset. The performance of the polynomial kernel over the two separate datasets supports the “No Free Launch Theorem”, which when applied to machine learning means that if an algorithm performs well over a class of problem(dataset), it may have worse performance on another class of problem or dataset. This implies that there might not be one specific machine learning algorithm that gives the best possible performance for a set of problems, it is therefore important for researchers to try out various algorithms before concluding on the best possible result on any dataset.

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